

7 Additional families of graphs

7.1 Asteroidal triple-free graphs

An *asteroidal triple* in a graph G is a set T of 3 vertices such that $G[T]$ is independent and every pair of vertices in T can be joined by a path that avoids the neighborhood of the third vertex in T . A graph G is *asteroidal triple-free* (AT-free) if G does not contain an asteroidal triple. Various classes of graphs are known to be AT-free, for example, interval graphs, co-comparability graphs and permutation graphs (see e.g. [COS97]). We show that all these classes have a bounded ratio by proving the following theorem.

Theorem 7.1. *For every asteroidal triple-free graph G , we have $\gamma(G) \leq 3 \cdot \rho(G)$.*

Proof. Our proof is based on the following property.

Theorem 7.2. [COS97] *Every connected asteroidal triple-free graph contains a pair of vertices such that any path between this pair of vertices is dominating.*

Let G be an AT-free graph. By Theorem 7.2 there is a pair of vertices $u, v \in V(G)$ such that any path between u and v is dominating. Let $\Pi = \pi_1, \pi_2, \dots, \pi_p$ be the shortest path between u and v . Thus, we have $\gamma(G) \leq p$. On the other hand as Π is a shortest path in G , by taking the set of vertices $\{\pi_{3i+1} \mid 0 \leq i \leq \lfloor p/3 \rfloor - 1\}$ we get a packing in G . Therefore, $\rho(G) \geq \lfloor p/3 \rfloor \geq \lfloor \gamma(G)/3 \rfloor$. $\square$

7.2 Convex graphs

A bipartite graph $G = (X \cup Y, E)$ is *convex* if and only if one of the color classes, without loss of generality X , has an ordering such that for all $y \in Y$ the vertices adjacent to y are consecutive in the ordering. We refer to this ordering as *convex ordering*.

Theorem 7.3. *For every convex graph G , we have $\gamma(G) \leq 3 \cdot \rho(G)$.*

Proof. We can assume that G does not have an isolated vertex. We will encode convex graphs using points and intervals in $\mathbb{R}$. In particular, if the convex ordering of X is $x_1, \dots, x_n$, we use the embedding $\pi : X \rightarrow \mathbb{R}$ defined as $\pi(x_i) = i$, and for each $y \in Y$, we associate the interval $\iota(y) = \{\pi(x) : x \in N_G(y)\}$, see Figure 4 for an illustration. For subsets $X' \subseteq X$ and $Y' \subseteq Y$, we use the notation $\pi(X') = \{\pi(x) : x \in X'\}$ and $\iota(Y') = \{\iota(y) : y \in Y'\}$. In this encoding, the image of a set $S \subseteq V(G)$ is a union of points and intervals, corresponding to the vertices in $S \cap X$ and $S \cap Y$, respectively.

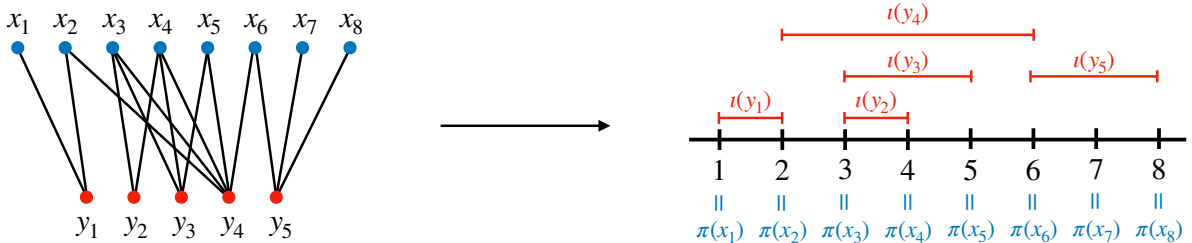


Figure 4: The embeddings π and ι .